

Power BI Assignment 2 – DAX & Data Visualization

Answers snips documents

Calculated Columns:

- Create a Calculated Column for 'Category Type': Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.
- Calculate Revenue per Order in Order Details Table: Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.
- Create a Calculated Column to Categorize Sales: Add a calculated column named 'Sales Category' in the Order Details table that categorizes each order as 'AboveAverage' or 'Below Average' based on the Amount value.

Answers below:

The screenshot shows the Power BI Desktop interface. The 'Column tools' ribbon is active, displaying options for Name, Data type, Format, Summarization, Data category, Sort by column, Data groups, Manage relationships, and New column. Below the ribbon, a data table is visible with columns: ID, Amount, Profit, Quantity, Category, Sub-Category, Profit Margin, Profit Status, category type, Revenue, and Sales category. The 'Revenue' column is calculated as 'Order Details'[Amount] * 'Order Details'[Quantity]. The 'Sales category' column is calculated based on the 'Amount' value, categorizing orders as 'Above average' or 'Below average'. The table contains 15 rows of data.

ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	category type	Revenue	Sales category
13	\$12	1	2	Clothing	Hankchief	0.0833333333333333	"Profit"	Clothing - Hankchief	\$24	Below average
8	\$257	23	5	Clothing	Hankchief	0.0894941634241245	"Profit"	Clothing - Hankchief	\$1,285	Below average
5	\$68	20	5	Clothing	Hankchief	0.294117647058824	"Profit"	Clothing - Hankchief	\$340	Below average
6	\$42	12	5	Clothing	Hankchief	0.285714285714286	"Profit"	Clothing - Hankchief	\$210	Below average
4	\$26	12	3	Clothing	Hankchief	0.461538461538462	"Profit"	Clothing - Hankchief	\$78	Below average
15	\$97	29	2	Clothing	Hankchief	0.298969072164948	"Profit"	Clothing - Hankchief	\$194	Below average
5	\$40	16	3	Clothing	Hankchief	0.4	"Profit"	Clothing - Hankchief	\$120	Below average
8	\$154	39	3	Clothing	Hankchief	0.253246753246753	"Profit"	Clothing - Hankchief	\$462	Below average
4	\$34	12	3	Clothing	Hankchief	0.352941176470588	"Profit"	Clothing - Hankchief	\$102	Below average
6	\$6	3	1	Clothing	Hankchief	0.5	"Profit"	Clothing - Hankchief	\$6	Below average
6	\$56	18	2	Clothing	Hankchief	0.321428571428571	"Profit"	Clothing - Hankchief	\$112	Below average
10	\$24	1	2	Clothing	Hankchief	0.0416666666666667	"Profit"	Clothing - Hankchief	\$48	Below average
10	\$14	2	1	Clothing	Hankchief	0.142857142857143	"Profit"	Clothing - Hankchief	\$14	Below average
10	\$18	8	2	Clothing	Hankchief	0.444444444444444	"Profit"	Clothing - Hankchief	\$36	Below average
1	\$32	7	3	Clothing	Hankchief	0.21875	"Profit"	Clothing - Hankchief	\$96	Below average
7	\$106	15	7	Clothing	Hankchief	0.141509433962264	"Profit"	Clothing - Hankchief	\$742	Below average
7	\$14	5	1	Clothing	Hankchief	0.357142857142857	"Profit"	Clothing - Hankchief	\$14	Below average
7	\$17	7	3	Clothing	Hankchief	0.411764705882353	"Profit"	Clothing - Hankchief	\$51	Below average
1	\$148	59	3	Clothing	Hankchief	0.398648648648649	"Profit"	Clothing - Hankchief	\$444	Below average
13	\$26	2	2	Clothing	Hankchief	0.0769230769230769	"Profit"	Clothing - Hankchief	\$52	Below average

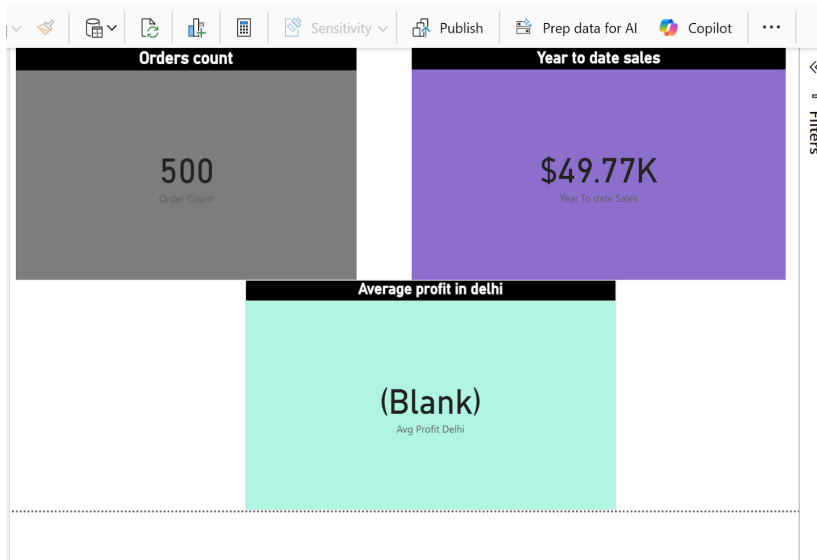
Calculated Measures:

- Calculate Order Count: Define a measure to count the total number of orders in the Order Details table.
- Calculate Average Profit in Delhi: Create a measure to calculate the average profit for orders placed in Delhi.
- Calculate Year-to-Date (YTD) Sales: Define a measure to calculate the total sales amount accumulated from the earliest order date up to each order date.

The screenshot shows a Power BI report interface. On the left, a table displays order details. The columns are Order ID, Order Date, CustomerName, and Locations. The table contains 20 rows of data, all from the 'Indore,Madhya Pradesh' location. On the right, the 'Data' pane shows a list of calculated measures. The measure 'Year To date Sales' is highlighted, and its DAX formula is displayed in the top pane: `1 Year To date Sales =`
`2 TOTALYTD (`
`3 SUM ( 'Order Details'[Amount] ),`
`4 'List of Orders'[Order Date]`
`5 )`

Order ID	Order Date	CustomerName	Locations
B-25627	4/23/2018	Hitika	Indore,Madhya Pradesh
B-25637	4/26/2018	Ashmi	Indore,Madhya Pradesh
B-25645	5/1/2018	Yasvi	Indore,Madhya Pradesh
B-25651	5/7/2018	Anurag	Indore,Madhya Pradesh
B-25655	5/11/2018	Nida	Indore,Madhya Pradesh
B-25663	5/19/2018	Pournamasi	Indore,Madhya Pradesh
B-25680	6/4/2018	Aayushi	Indore,Madhya Pradesh
B-25681	6/4/2018	Bhawna	Indore,Madhya Pradesh
B-25685	6/10/2018	Sheetal	Indore,Madhya Pradesh
B-25688	6/11/2018	Saetha	Indore,Madhya Pradesh
B-25690	6/15/2018	Gurjan	Indore,Madhya Pradesh
B-25692	6/17/2018	Rashmi	Indore,Madhya Pradesh
B-25701	6/26/2018	Maithilee	Indore,Madhya Pradesh
B-25703	6/26/2018	Eita	Indore,Madhya Pradesh
B-25705	6/30/2018	Shweta	Indore,Madhya Pradesh
B-25708	7/1/2018	Kishwar	Indore,Madhya Pradesh
B-25709	7/1/2018	Aakarika	Indore,Madhya Pradesh
B-25719	7/12/2018	Rashmi	Indore,Madhya Pradesh
B-25724	7/19/2018	Sheetal	Indore,Madhya Pradesh
B-25727	7/22/2018	Turumella	Indore,Madhya Pradesh

Report view of calculated Measures:

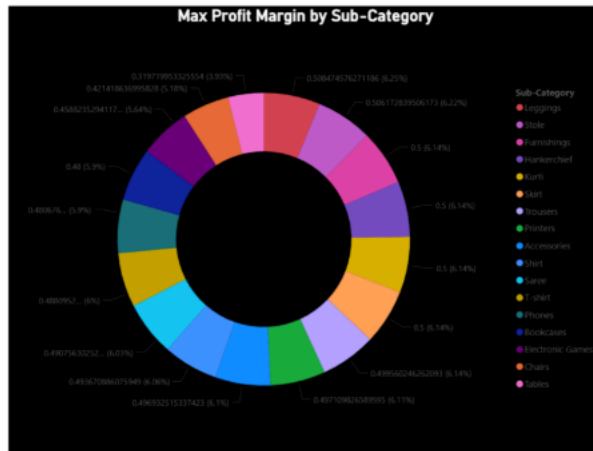


Data Visualization: Answers below:

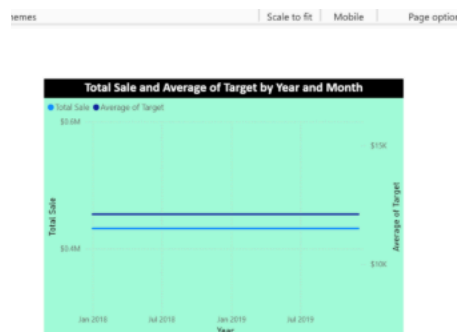
1.Sales Target Achievement by Category: Compare actual sales with sales targets by category using a clustered column chart.



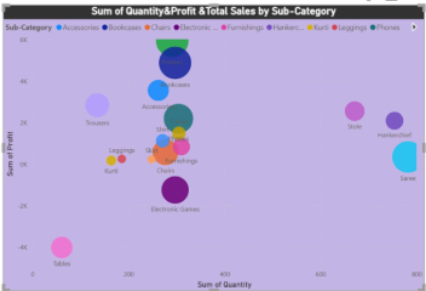
2. Max Profit Margin by Sub-Category: Analyze the maximum profit margin for each sub-category of products using a donut chart.



3. Monthly Sales Trend: Show the trend of monthly sales over time using a line chart.



profit and quantity sold for different sub-categories using a scatter chart.



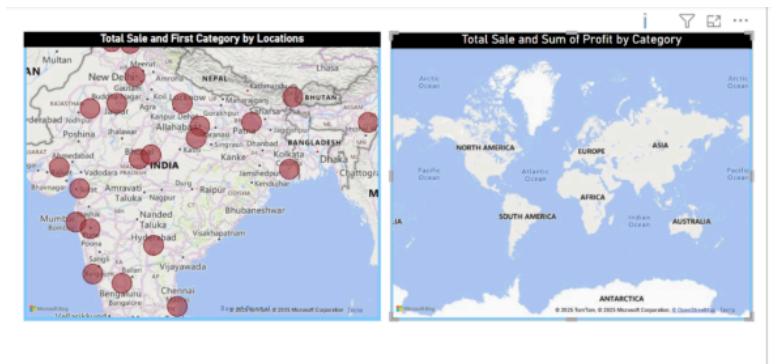
create a multi-row card to display the minimum target for each segment.



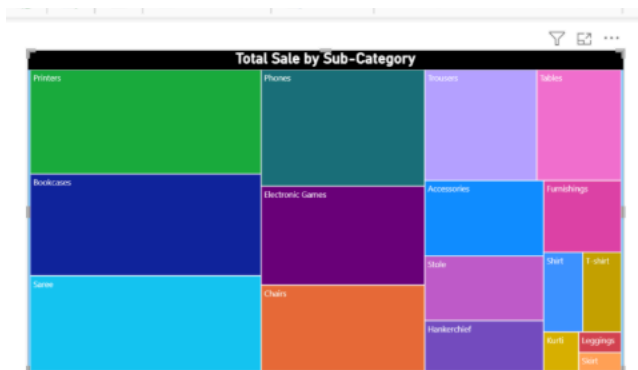
to sales targets across different categories and months.

Sales Performance														
Month	January		February		March		April		May		June		July	
Category	Total Sales	Sum of Target	Total Sales	Sum of Target	Total Sales	Sum of Target	Total Sales	Sum of Target	Total Sales	Sum of Target	Total Sales	Sum of Target	Total Sales	Sum of Target
Clothing	\$139,054	\$139,000	\$139,054	\$139,000	\$139,054	\$139,000	\$139,054	\$139,000	\$139,054	\$139,000	\$139,054	\$139,000	\$139,054	\$139,000
Electronics	\$165,267	\$165,000	\$165,267	\$165,000	\$165,267	\$165,000	\$165,267	\$165,000	\$165,267	\$165,000	\$165,267	\$165,000	\$165,267	\$165,000
Furniture	\$127,811	\$127,000	\$127,811	\$127,000	\$127,811	\$127,000	\$127,811	\$127,000	\$127,811	\$127,000	\$127,811	\$127,000	\$127,811	\$127,000
Total	\$431,990	\$431,000	\$431,990	\$431,000	\$431,990	\$431,000	\$431,990	\$431,000	\$431,990	\$431,000	\$431,990	\$431,000	\$431,990	\$431,000

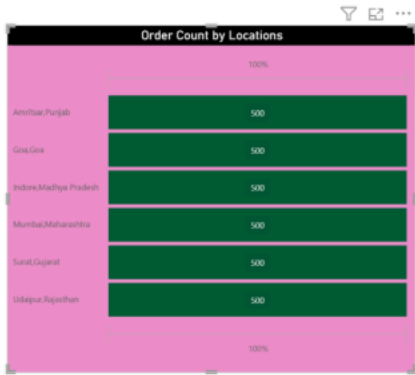
7. Geographic Sales Analysis: Visualize total sales on a map by city to identify regional sales patterns.



8. Sales Distribution by Sub-Category: Represent the sales distribution across different sub-categories using a tree map.



9. Order Count Analysis by State: Create a funnel chart to visualize the distribution of order counts across different states.



Thank you

Valli.